OPERATING SYSTEMS



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WHAT IS AN OPERATING SYSTEM?



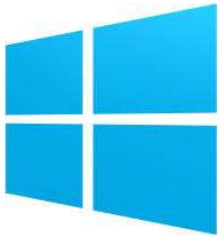
WHAT IS AN OPERATING SYSTEM?



Linux



macOS X
iOS



Microsoft Windows



Android



WHAT IS AN OPERATING SYSTEM?

- **An Operating System (OS) is a well-organized collection of programs that manages the computer hardware.**
- **It acts as an intermediary between the user and the computer hardware**
- **The OS coordinates multiple applications and users**
- **An OS provides standard services and coordination.**



OPERATING SYSTEM GOALS

- **Convenience**
- **Efficiency**
- **Ability to evolve**



HOW TO CHOOSE THE CORRECT OS?

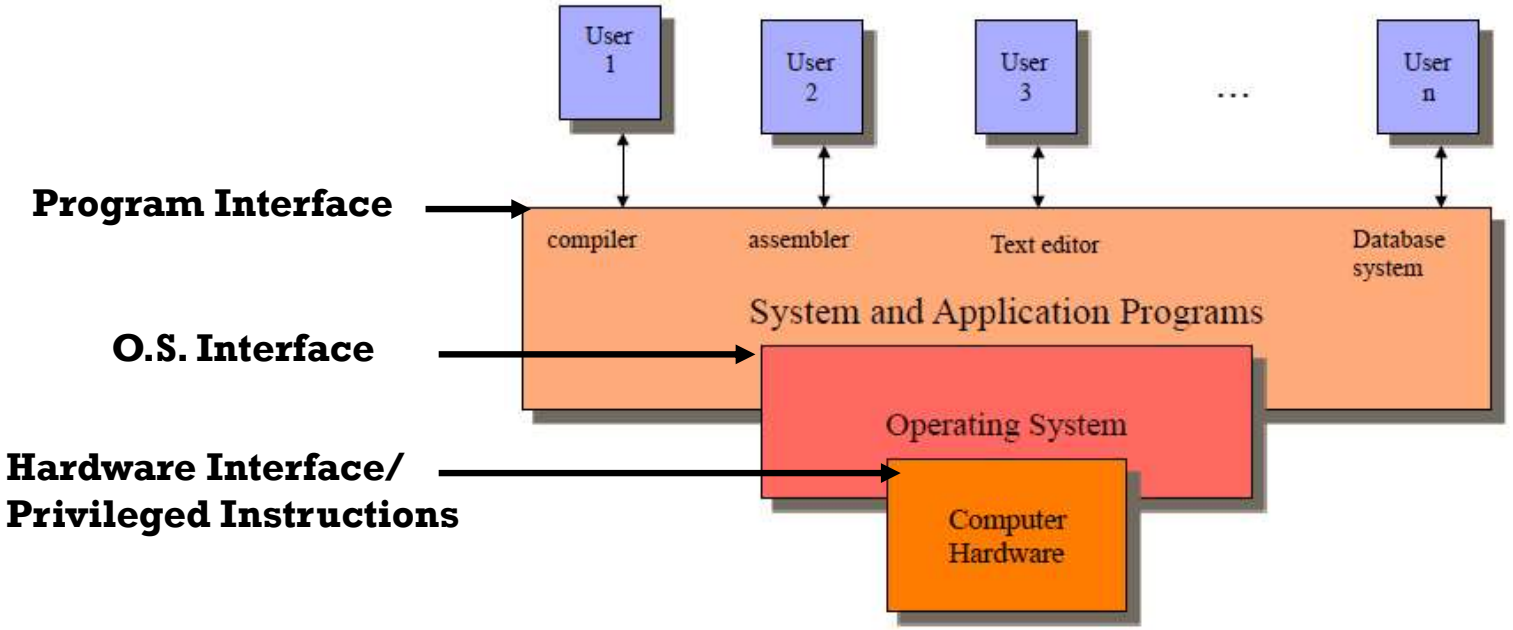


HOW TO CHOOSE THE CORRECT OS?

- Price
- Accessibility
- Compatibility
- Security



COMPUTER SYSTEM STRUCTURE



OPERATING SYSTEM ROLES

- **Referee**
- **Illusionist**
- **Glue**



WHY SHOULD I STUDY OPERATING SYSTEMS?



SYSTEMS TODAY







OPERATING SYSTEM SPECTRUM

- **Batch and multiprogramming**
- **Timesharing (multitasking)**
- **Personal Computing**
- **Parallel and Distributed**
- **Real Time**



COMPUTER STARTUP

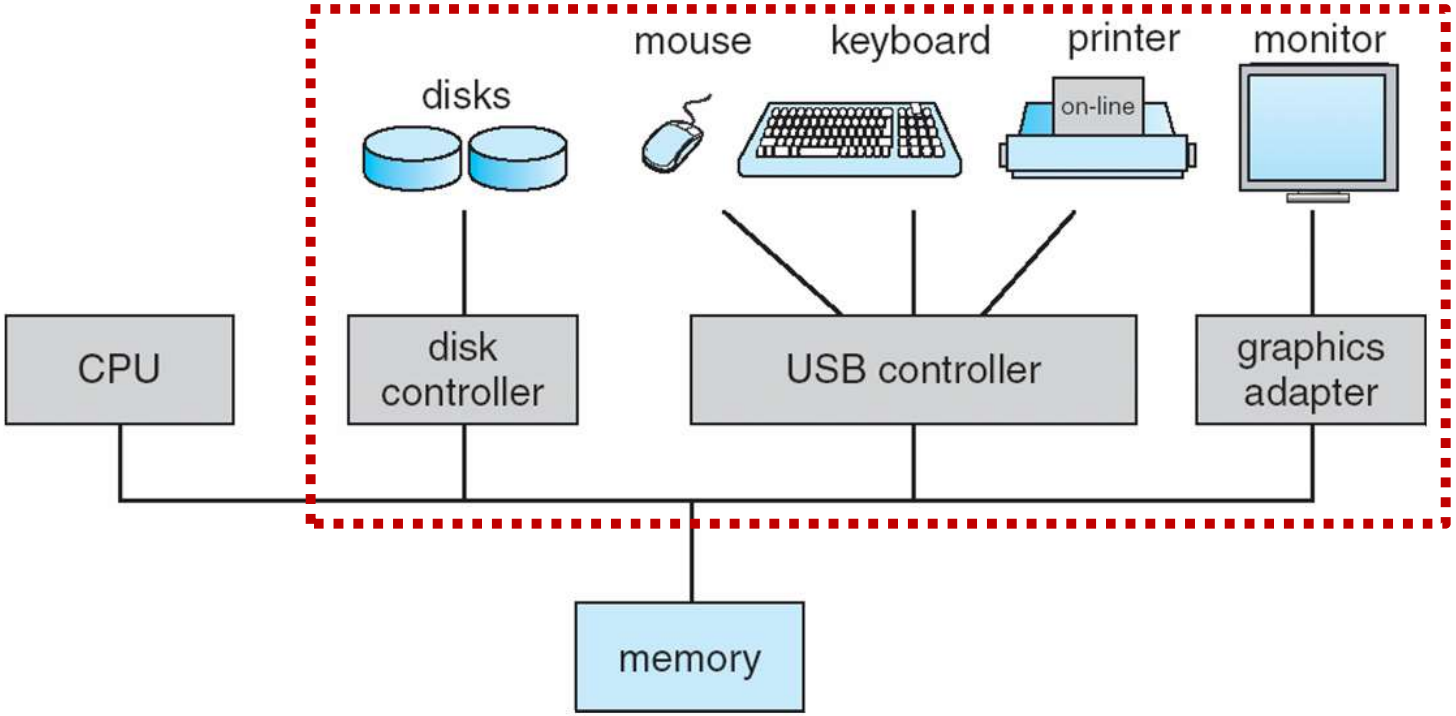
- **Bootstrap program is loaded at power-up or reboot**
  - **stored in ROM (firmware)**
  - **Initializes all aspects of system**
  - **Loads operating system kernel and starts execution**



OVERVIEW OF COMPUTER SYSTEM STRUCTURE



COMPUTER SYSTEM ORGANIZATION



COMPUTER SYSTEM ORGANIZATION

- **A large portion of operating system code is dedicated to managing I/O.**
- **I/O devices and the CPU can execute concurrently**
- **Device controller is in charge of a particular device type**
- **OS have a device driver for each device controller.**
- **Device controller informs CPU that it has finished its operation by causing an interrupt**

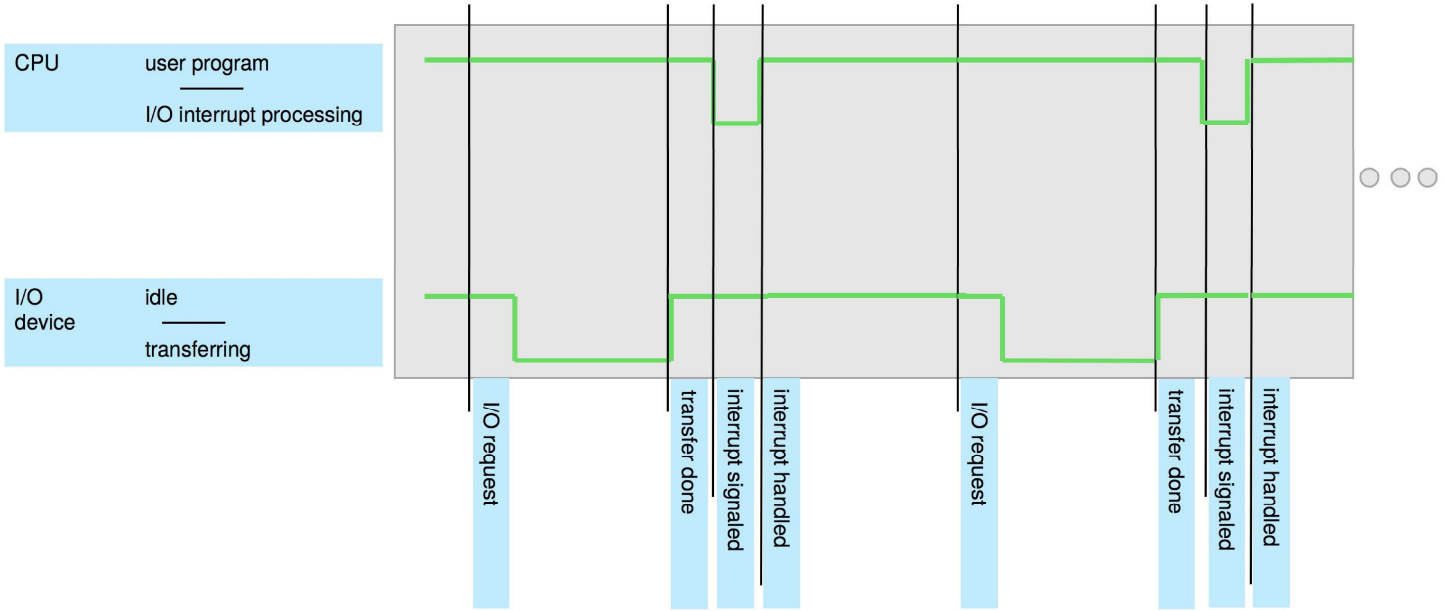


INTERRUPTS

- **Signal emitted by hardware or software when a process or an event needs immediate attention.**
- **It alerts the processor to a high-priority process requiring interruption of the current working process.**
- **Hardware may trigger an interrupt at the any time by the sending a signal to the CPU**
- **When CPU interrupted, it is stop what it is doing and immediately transfers the controller to the Interrupt Service Routine (ISR).**



INTERRUPT TIMELINE



TYPES OF INTERRUPT

- **Hardware interrupts**
- **Software interrupts**

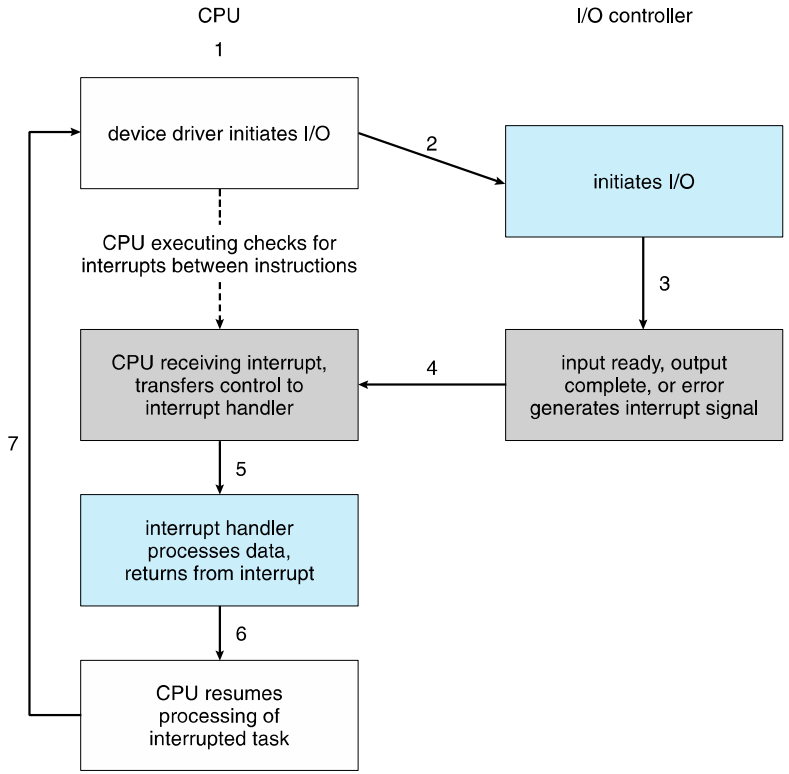


INTERRUPT HANDLING

- **OS preserves the state of the CPU by storing the registers and the program counter**
- **Determines which type of interrupt has occurred:**
- **Separate segments of code determine what action should be taken for each type of interrupt**



INTERRUPT-DRIVE I/O CYCLE

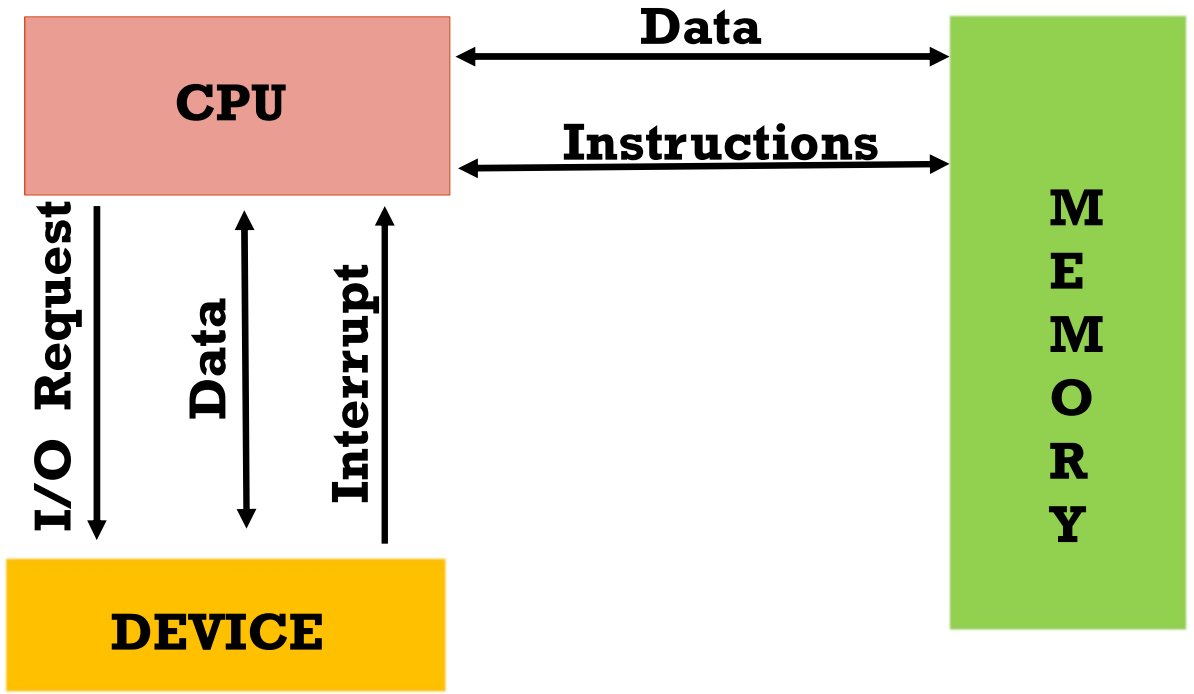


I/O STRUCTURE

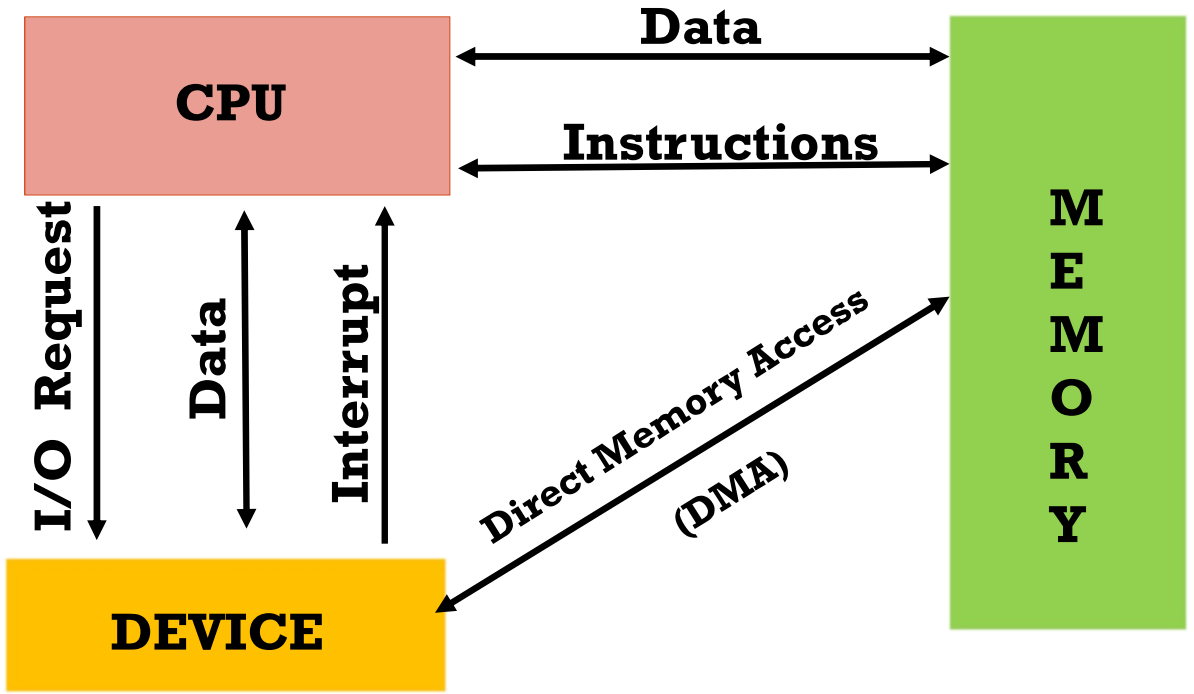
- **After I/O starts,**
  - **Control returns to user program only upon I/O completion**
  - **Control returns to user program without waiting for I/O completion**



HOW A MODERN COMPUTER WORKS



HOW A MODERN COMPUTER WORKS

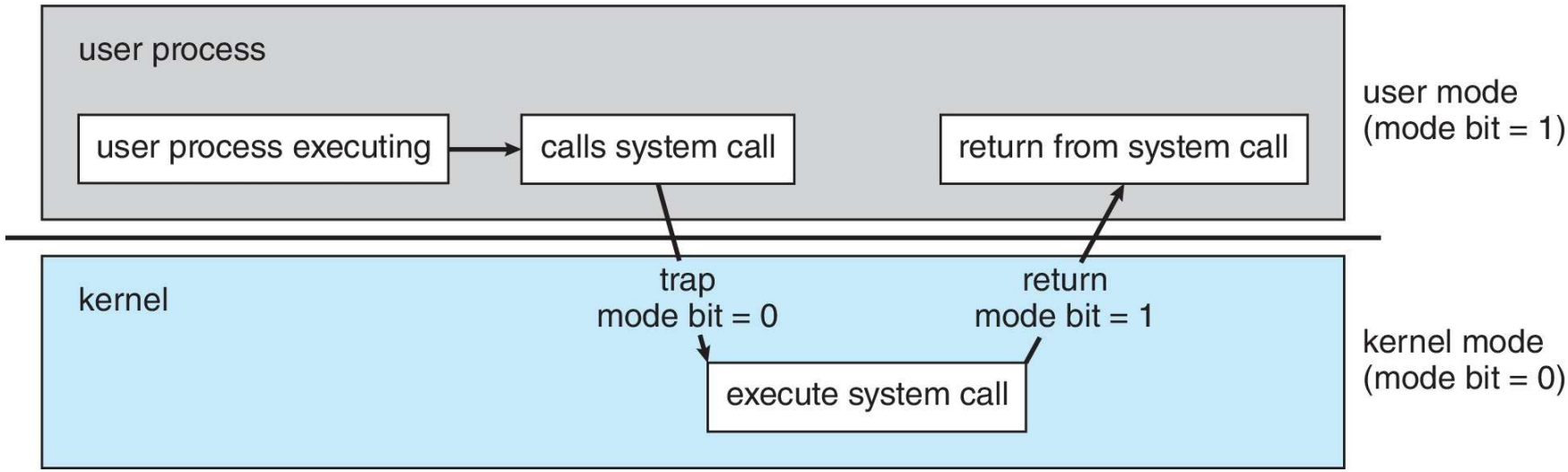


DUAL-MODE OPERATION

- Allows OS to protect itself and other system components
  - User mode
  - kernel mode
- Mode bit (provided by hardware)
- Provides ability to distinguish when system is running user code or kernel code.
  - user is running - mode bit is "user"
  - kernel code is executing - mode bit is "kernel"
- System call changes mode to kernel, return from call resets it to user
- Some instructions designated as privileged, only executable in kernel mode



TRANSITION FROM USER TO KERNEL MODE



STORAGE STRUCTURE

- **Main memory**
- **Secondary storage**
- **Solid-state disks**
  - **faster than hard disks,**
  - **nonvolatile**

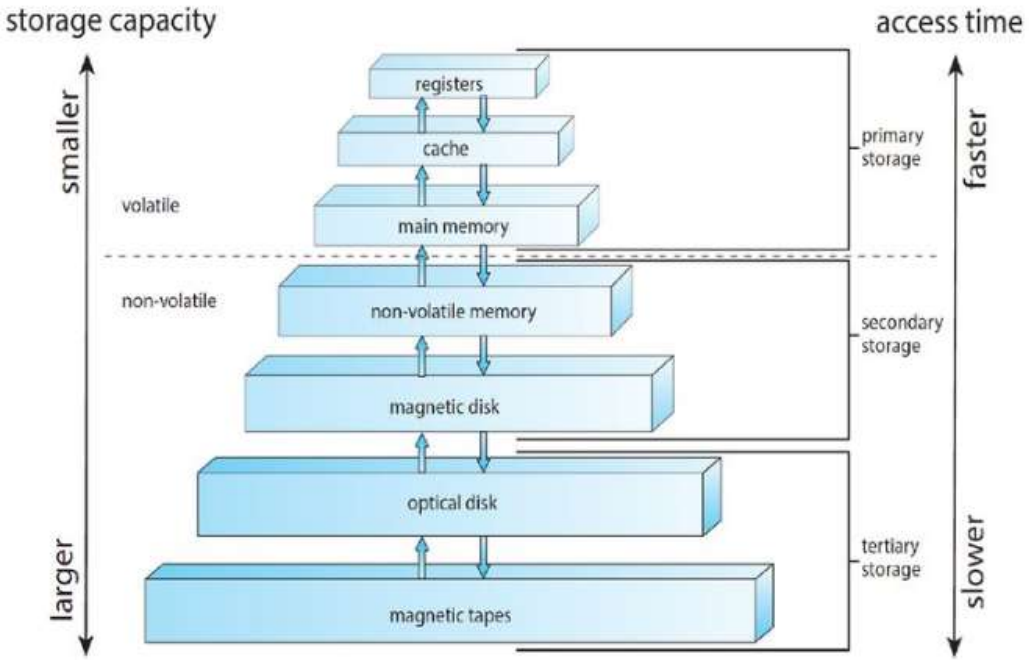


STORAGE HIERARCHY

- **Storage systems organized in hierarchy**
  - **Speed**
  - **Cost**
  - **Volatility**
- **Main memory can be viewed as a cache for secondary storage**



STORAGE-DEVICE HIERARCHY



TIMER

- **Prevent infinite loop (or process hogging resources)**
- **Set to interrupt the computer after some time period**
- **Keep a counter that is decremented by the physical clock**
- **Operating system set the counter (privileged instruction)**
- **When counter zero generate an interrupt**
- **Set up before scheduling process to regain control or terminate program that exceeds allotted time**



PROCESS MANAGEMENT

- A process is a program in execution. It is a unit of work within the system.
- Program is a passive entity, process is an active entity.
- Process needs resources to accomplish its task
  - CPU, memory, I/O, files
  - Initialization data
- Process termination requires reclaim of any reusable resources
- Single-threaded process has one program counter
- Multi-threaded process has one program counter per thread
- Typically system has many processes,
  - Maintain the concurrency by multiplexing the CPUs among the processes / threads



PROCESS MANAGEMENT ACTIVITIES

- **Creating and deleting both user and system processes**
- **Suspending and resuming processes**
- **Providing mechanisms for**
  - **Process synchronization**
  - **Process communication**
  - **Deadlock handling**



MEMORY MANAGEMENT

- **Program Instructions and data that is needed by the program are in the memory**
- **Determines what is in memory and when Optimizing CPU utilization and computer response to users**
- **Memory management activities**
  - **Keeping track of which parts of memory are currently being used and by whom**
  - **Deciding which processes (or parts thereof) and data to move into and out of memory**
  - **Allocating and de-allocating memory space as needed**



FILE-SYSTEM MANAGEMENT

- **OS provides uniform, logical view of information storage**
  - **Abstracts physical properties to logical storage unit - file**
  - **Each medium is controlled by device (i.e., disk drive, tape drive)**
    - **Varying properties include access speed, capacity, data-transfer rate, access method (sequential or random)**



FILE-SYSTEM MANAGEMENT

- **Files usually organized into directories**
- **Access control on most systems to determine who can access it and level of access**
- **OS activities include**
  - **Creating and deleting files and directories**
  - **Primitives to manipulate files and directories**
  - **Mapping files onto secondary storage**
  - **Backup files onto stable (non-volatile) storage media**



MASS-STORAGE MANAGEMENT

- **Usually disks used to store data**
- **It may not fit in main memory or data**
- **Entire speed of computer operation hinges on disk subsystem and its algorithms**
- **OS activities**
  - **Free-space management**
  - **Storage allocation**
  - **Disk scheduling**
- **Some storage need not be fast**
  - **Tertiary storage includes optical storage, magnetic tape**



CACHING

- **Use copied from slower to faster storage temporarily**
- **Faster storage (cache) checked first to determine if information is there**
  - **If it is, directly from the cache**
  - **If not, data copied to cache and used**



PERFORMANCE OF VARIOUS LEVELS OF STORAGE

Level	1	2	3	4	5
Name	registers	cache	main memory	solid state disk	magnetic disk
Typical size	< 1 KB	< 16MB	< 64GB	< 1 TB	< 10 TB
Implementation technology	custom memory with multiple ports CMOS	on-chip or off-chip CMOS SRAM	CMOS SRAM	flash memory	magnetic disk
Access time (ns)	0.25 - 0.5	0.5 - 25	80 - 250	25,000 - 50,000	5,000,000
Bandwidth (MB/sec)	20,000 - 100,000	5,000 - 10,000	1,000 - 5,000	500	20 - 150
Managed by	compiler	hardware	operating system	operating system	operating system
Backed by	cache	main memory	disk	disk	disk or tape



I/O SUBSYSTEM

- **I/O subsystem responsible for**
  - **Memory management of I/O including**
    - **Buffering**
    - **caching**
    - **spooling**
  - **General device-driver interface (Drivers has specific hardware devices)**
  - **Drivers for specific hardware devices**



PROTECTION AND SECURITY

- **Protection –mechanism for controlling access of processes or users to resources defined by the OS**
- **Security – defense of the system against internal and external attacks**
  - **Huge range, including denial-of-service, worms, viruses, identity theft, theft of service**
- **Systems generally first distinguish among users, to determine who can do what**
  - **User identities (user IDs, security IDs) ,one per user**
  - **User ID then associated with all files, processes of that user to determine access control**
  - **Group identifier (group ID)**
  - **Privilege escalation allows user to change to effective ID with more rights**

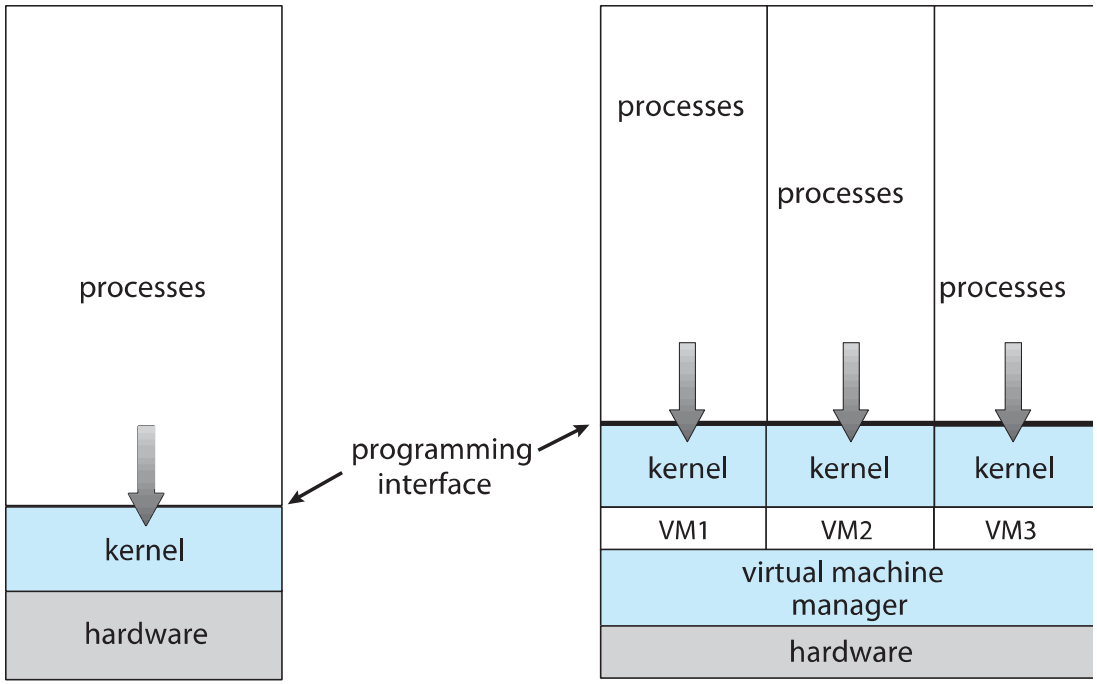


VIRTUALIZATION

- **Allows operating systems to run applications within other OSes**
- **Emulation used when source CPU type different from target type**
- **Use cases involve laptops and desktops running multiple OSes for exploration or compatibility**
- **VMM can run natively, in which case they are also the host**



VIRTUALIZATION (COMPUTING ENVIRONMENTS)



COMPUTER SYSTEM ARCHITECTURE



COMPUTER-SYSTEM ARCHITECTURE

- **Single processor Systems**
- **Multiprocessors Systems**
- **Clustered Systems**



COMPUTER-SYSTEM ARCHITECTURE

- **Single processor Systems**
  - **One CPU execute general purpose instruction set**



COMPUTER-SYSTEM ARCHITECTURE

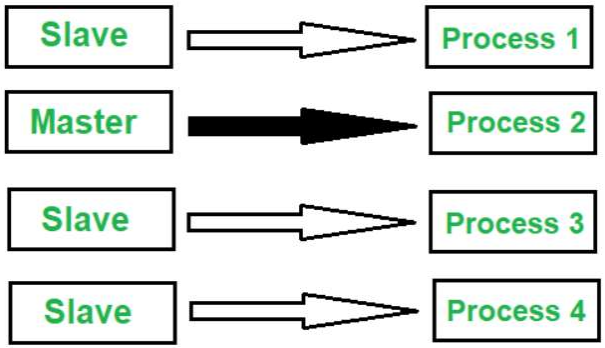
- **Multiprocessors Systems**
  - **Two or more central processing units within a single computer system**
  - **Sharing computer bus, clock, memory, peripheral devices**
  - **We call as parallel systems or tightly-coupled systems**

- **Advantages of Multiprocessors systems**
  1. **Increased throughput**
  2. **Economy of scale**
  3. **Increased reliability**

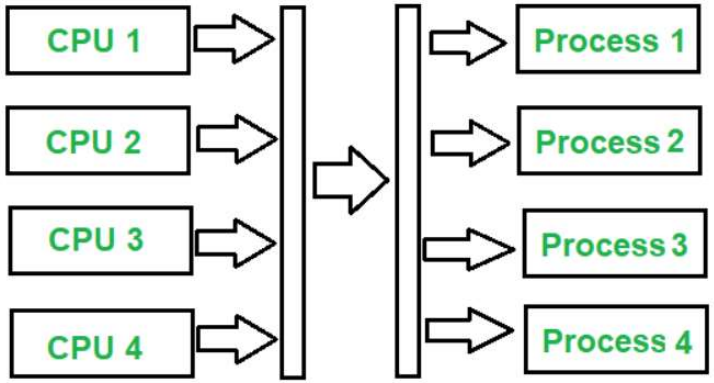


TYPES OF MULTIPROCESSORS SYSTEMS

Asymmetric Multiprocessing



Symmetric Multiprocessing



TYPES OF MULTIPROCESSORS SYSTEMS

- **Difference Between Asymmetric and Symmetric Multiprocessing**

Asymmetric	Symmetric
Use master-slave method and processors are not treated equally	All processors are treated equally and process is taken from the ready queue
Tasks of the operating system are done by master processor	Tasks of the operating system are done individual processor
No Communication between processors as they are controlled by the master processor	All processors communicate with another processor by a shared memory
Cheaper and easier to design	Cost and complex to design

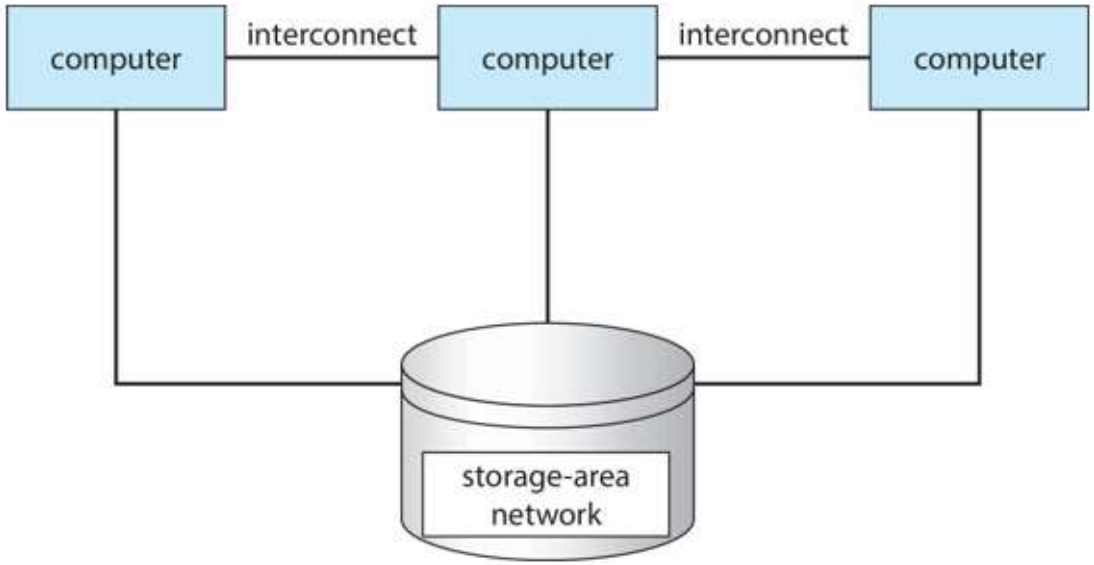


CLUSTERED SYSTEMS

- Like multiprocessor systems, but multiple systems working together
- Sharing storage via a storage-area network
- Provides a high-availability service
  - Asymmetric clustering
  - Symmetric clustering
- Some are for high-performance computing (HPC) clusters
  - Applications must be written to use parallelization
- Distributed lock manager (DLM) is used to avoid conflicting operations



CLUSTERED SYSTEMS



COMPUTING ENVIRONMENTS



COMPUTING ENVIRONMENTS

- **Traditional**
- **Mobile**
- **Client Server**
- **Peer-to-Peer**
- **Cloud computing**
- **Real-time Embedded**



COMPUTING ENVIRONMENTS (TRADITIONAL)

- **Stand-alone general purpose machines**
- **most systems interconnect with others (i.e., the Internet)**
- **Portals provide web access to internal systems**
- **Network computers (thin clients) are like Web terminals**
- **Mobile computers interconnect via wireless networks**



COMPUTING ENVIRONMENTS - MOBILE

- **Handheld smartphones, tablets, etc..**

- **What is the functional difference between them and a "traditional" laptop?**
  - **Extra feature – more OS features (GPS, gyroscope)**
  - **Allows new types of apps like augmented reality**
  - **Use IEEE 802.11 wireless, or cellular data networks for connectivity**



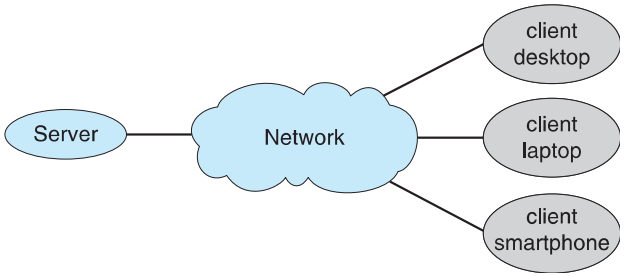
COMPUTING ENVIRONMENTS (DISTRIBUTED)

- **Collection of separate, heterogeneous, systems networked together**
  - **Network is a communications path, TCP/IP most common**
    - **Local Area Network (LAN)**
    - **Wide Area Network (WAN)**
    - **Metropolitan Area Network (MAN)**
    - **Personal Area Network (PAN)**
- **Network Operating System provides features between systems across network**
  - **Communication scheme allows systems to exchange messages Illusion of a single system**



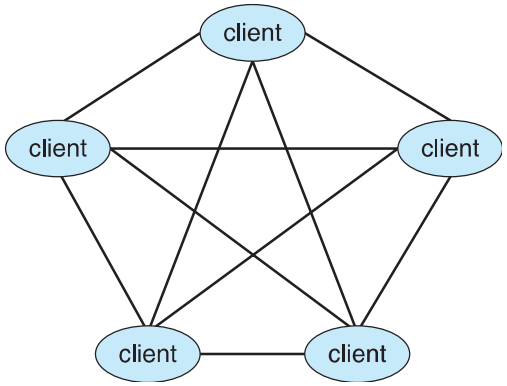
COMPUTING ENVIRONMENTS (CLIENT-SERVER)

- **Dumb terminals supplanted by smart PCs**
- **Many systems now servers, responding to requests generated by clients**
  - **Compute-server system provides an interface to client to request services (i.e., database)**
  - **File-server system provides interface for clients to store and retrieve files**



COMPUTING ENVIRONMENTS — (PEER-TO-PEER)

- **Another model of distributed system**
- **P2P does not distinguish clients and servers**
  - **Instead all nodes are considered peers**
  - **May each act as client, server or both**
  - **Node must join P2P network**
- **Examples :Voice over IP (VoIP) such as Skype**



COMPUTING ENVIRONMENTS (CLOUD COMPUTING)

- **Delivers computing, storage, even apps as a service across a network**
- **Logical extension of virtualization because it uses virtualization as the base for it functionality.**
  - **Amazon EC2 has thousands of servers, millions of virtual machines, petabytes of storage available across the Internet, pay based on usage**
- **Cloud computing environments composed of traditional OSes, plus VMMs, plus cloud management tools**
  - **Internet connectivity requires security like firewalls**
  - **Load balancers spread traffic across multiple applications**



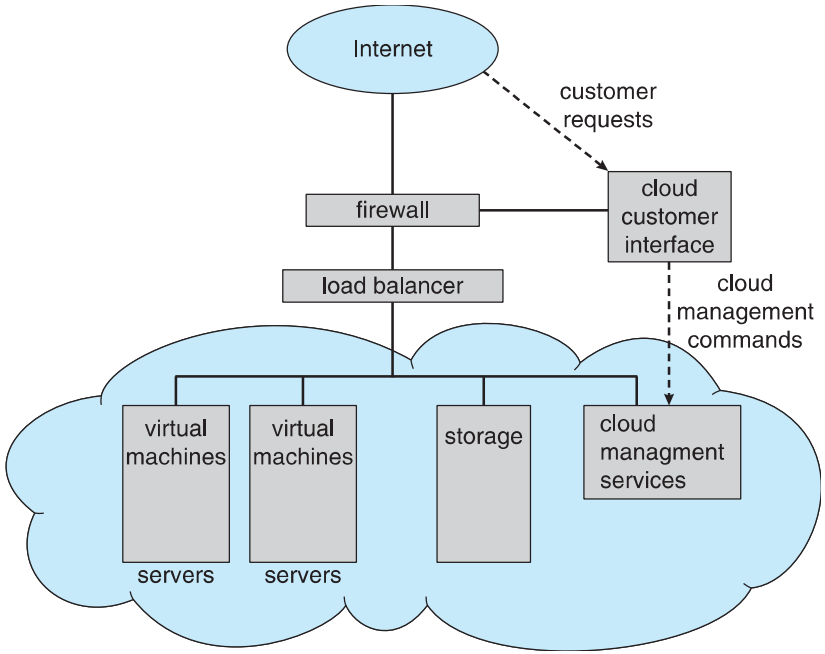
COMPUTING ENVIRONMENTS (CLOUD COMPUTING)

- **Types**
  - **Public cloud - available via Internet to anyone willing to pay**
  - **Private cloud (for the company's own use)**
  - **Hybrid cloud – (public and private cloud components)**

  - **Software as a Service (SaaS)**
  - **Platform as a Service (PaaS)**
  - **Infrastructure as a Service (IaaS)**



COMPUTING ENVIRONMENTS (CLOUD COMPUTING)



COMPUTING ENVIRONMENTS – (REAL-TIME EMBEDDED SYSTEMS)

- **Real-time embedded systems most prevalent form of computers**
  - **Vary considerable, special purpose, limited purpose OS, real-time OS**
- **Many other special computing environments as well**
  - **Some have OSes, some perform tasks without an OS**
- **Real-time OS has well-defined fixed time constraints**
  - **Processing must be done within constraint**
  - **Correct operation only if constraints met**

